AI-Powered Job Application Automation Platform

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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CHAPTER 1. INTRODUCTION

1.1.Client Identification, Need Identification and Relevant Contemporary issue

This section focuses on identifying the target users of the system, the key challenges they face, and the contemporary relevance of the problem. By examining current recruitment practices and their limitations, this section justifies the need for the proposed solution and establishes the context for the project.

1.1.1 Client Identification

The primary clients of the proposed system are students, fresh graduates, and early-career professionals who are actively seeking employment in an increasingly automated and technology-driven recruitment environment. These users typically interact with multiple digital platforms such as job portals, professional networking sites, institutional ERP systems, and company-specific application portals during their job search.

In addition to individual job seekers, the system indirectly benefits:

- placement cells and academic institutions seeking structured career data,
- career mentors and advisors who require holistic visibility into a candidate's profile, and
- employers who rely on structured, ATS-compatible candidate information.

The project is therefore designed with a candidate-centric focus, addressing the real-world challenges faced by individuals entering or navigating the modern job market.

1.1.2 Need Identification

The need for this project arises from the inefficiency, redundancy, and fragmentation inherent in current job application workflows. Job seekers are required to repeatedly enter the same personal and professional information across different platforms, often in slightly different formats.

Key needs identified include:

- A single centralized repository for storing complete professional identity data.
- The ability to reuse career information across multiple job applications without repetitive manual entry.
- Support for role-specific customization of resumes and application content.
- Improved alignment with Applicant Tracking Systems (ATS) to prevent premature rejection.
- Transparency and user control over automated or AI-assisted processes.

Preliminary observations and informal surveys among university seniors indicate that a significant majority of candidates experience "application fatigue" and desire a system that can intelligently adapt their career data to different job contexts while maintaining accuracy and trust.

1.1.3 Identification of Relevant Contemporary Issue

A major contemporary issue addressed by this project is **Career Data Fragmentation**. In the current professional ecosystem, a candidate's career identity is scattered across multiple disconnected platforms, including:

- LinkedIn profiles

- locally stored resumes
- personal portfolio websites
- university ERP systems
- company-specific job portals

Each platform maintains its own representation of the candidate, leading to inconsistencies, outdated information, and duplication of effort. There is no standardized mechanism to unify, validate, and reuse this data across systems

1.1.4 Justification of the Contemporary Issue

The relevance and severity of career data fragmentation are supported by multiple sources:

- ATS Filtering Statistics: Industry reports suggest that nearly 75% of resumes are filtered out by Applicant Tracking Systems before reaching human recruiters, often due to poor keyword alignment or formatting inconsistencies.

- Application Fatigue: A typical job application requires candidates to manually complete 16 or more distinct sections, such as education, experience, projects, certifications, and volunteer work, for each employer portal.

- User Survey Insights: Preliminary surveys among final-year students reveal that 85% find existing platforms insufficient for role-specific tailoring, while 90% express the need for a centralized "single source of truth" for their career data.

- Global Workforce Reports: Organizations such as the World Economic Forum highlight the growing skills gap and emphasize the need for more granular, continuously updated, and validated career profiles to improve labor market efficiency.

These factors collectively establish career data fragmentation as a significant contemporary issue, warranting a systematic and technology-driven solution.

1.2. Identification of Problem

Despite advancements in recruitment technology, there is no unified system that effectively manages a candidate's professional identity as a reusable and evolving asset. Existing processes are largely manual, repetitive, and static, lacking the intelligence required to map a candidate's comprehensive skill set to the diverse and dynamic requirements of global job descriptions (JDs).

The absence of such a system leads to:

- missed opportunities for qualified candidates,
- inefficient job matching,
- prolonged hiring cycles for employers, and
- reduced trust in automated recruitment tools.

Problem addressed by this project is the lack of a centralized and intelligent platform for managing professional identity, capable of bridging the gap b/w candidate profiles and role-specific job requirements.

1.3. Phase-wise Task Identification and Development Plan

This project follows a structured, phase-wise development approach to ensure systematic progress, modular implementation, and continuous validation of outcomes. Each phase addresses a specific layer of the system, starting from secure identity management and progressing toward intelligent automation and deployment. The phased plan ensures that foundational components are stabilized before introducing advanced intelligence and automation features.

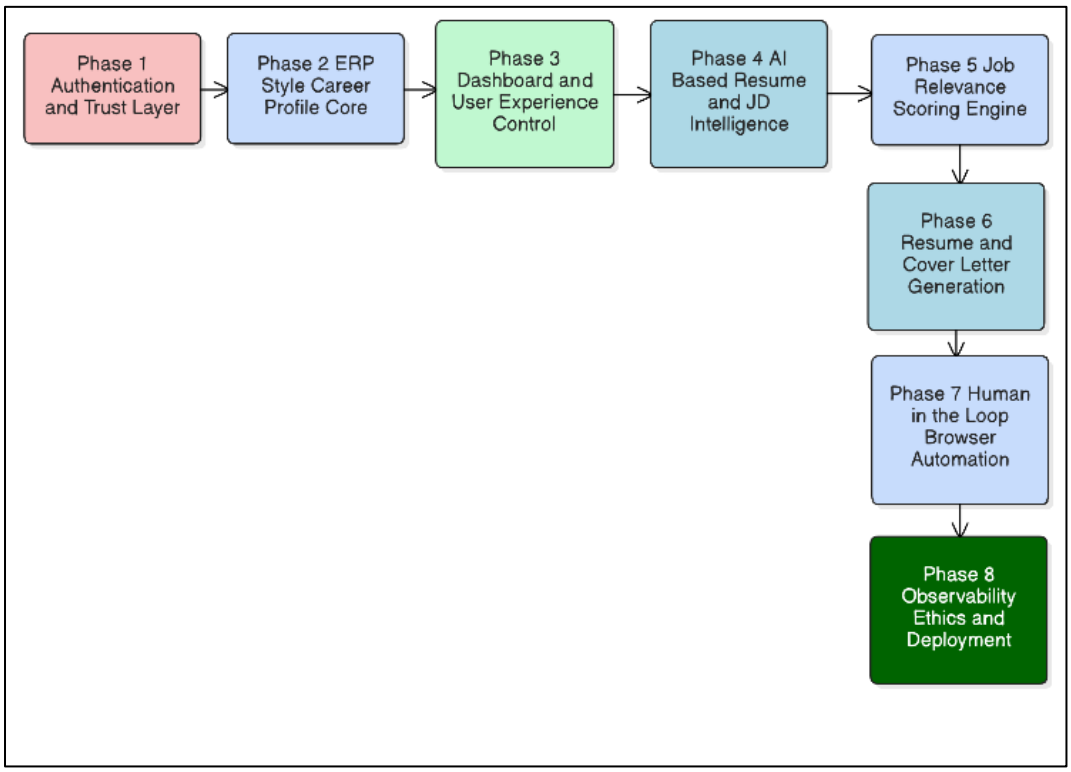

Figure 1 Phase-wise Development flow

- **Phase 1: Authentication and Trust Layer** The first phase focuses on establishing a secure and reliable identity management system, which forms the foundation of the entire platform. Since all subsequent modules career profile management, AI-based analysis, and automation depend on verified user identities, this phase prioritizes security, correctness, and trust. During this phase, the system implements email–password authentication with secure password hashing, JWT-based session management, refresh token handling, and third-party authentication using Google OAuth 2.0. Email-based OTP verification is integrated to ensure account authenticity, and access to the platform is restricted until identity verification is complete. Corresponding frontend interfaces for signup, login, OTP verification, and protected route handling are developed to ensure a seamless yet secure user experience.

  The successful completion of this phase ensures that users can securely register, authenticate, verify their identity, and manage sessions within the platform.

- **Phase 2: ERP-Style Career Profile Core** The second phase establishes the core Career ERP system, which serves as a single source of truth for a user's professional identity. The focus of this phase is to design a flexible yet structured data model capable of representing comprehensive career information in a reusable and auditable manner. This phase introduces a versioned career profile schema, clearly distinguishing mandatory and optional fields. Users can construct their profiles through manual entry, resume uploads, or LinkedIn PDF ingestion, all of which are normalized into a unified schema. AI-

extracted data is presented for manual review to ensure correctness before persistence. Additionally, profile completeness is computed at both field and section levels to provide transparent progress feedback.

At the end of this phase, each user possesses a complete, versioned, and verifiable career identity profile.

- **Phase 3: Dashboard and User Experience Control** Phase three focuses on usability, transparency, and control, achieved through a centralized dashboard interface. This phase ensures that users can easily understand and manage their professional data without being overwhelmed by raw information. The dashboard retrieves the active profile version, computes real-time profile completion metrics, and displays the most recent update timestamps. Users are provided with clear actions to edit their profiles or securely log out. Profile updates trigger the creation of new profile versions while preserving historical data.

  Upon completion, users have full visibility and control over their career identity within the platform.

- **Phase 4: AI-Based Resume and Job Description Intelligence** The fourth phase introduces semantic intelligence into the system by enabling AI-driven understanding of candidate profiles and job descriptions. This phase focuses on interpreting unstructured textual data rather than executing actions. AI models are employed to extract skills, experiences, and projects from resumes, distinguish between explicit and implied skills, and parse job descriptions to identify role expectations and experience levels. The system then performs skill confidence mapping and generates explainable fit and gap analyses.

  This phase enables the platform to understand both candidates and jobs at a semantic level.

- **Phase 5: Job Relevance Scoring Engine** Phase five builds the decision-support layer of the system. Using both rule-based logic and LLM-assisted contextual reasoning, the platform evaluates candidate–job compatibility. Relevance scores are computed through weighted aggregation of skill overlap, experience alignment, and contextual fit. Thresholds are defined to interpret relevance levels, and detailed score breakdowns are generated to maintain transparency.

  Successful completion allows the system to rank job opportunities intelligently and explainable.

- **Phase 6: Resume and Cover Letter Generation** The sixth phase converts analytical intelligence into actionable outputs. Based on job relevance analysis, the platform generates role-specific resume variants and personalized cover letters optimized for ATS compatibility. Users are given full control to preview, edit, regenerate, or approve generated documents, ensuring trust and usability.

  This phase ensures that AI assistance enhances, rather than replaces, user decision-making.

- **Phase 7: Human-in-the-Loop Browser Automation** Phase seven introduces controlled automation to assist users in submitting job applications. Automation is implemented using a human-in-the-loop model to prevent blind execution. The system uses browser automation to fill forms and upload documents, while pausing execution when encountering captchas, two-factor authentication, or unexpected inputs. Failures are logged with screenshots and retry logic to maintain reliability and auditability.

  This phase ensures applications are executed safely with explicit user oversight.

- **Phase 8: Observability, Ethics, and Deployment** The final phase focuses on making the system production-ready and research-defensible. Centralized logging and audit trails are implemented to track authentication events, profile updates, and automation attempts. User consent mechanisms and automation disclosures are introduced to address ethical considerations. The platform is containerized and deployed on cloud infrastructure, followed by the preparation of demonstrations, documentation, and research

artifacts.

At the end of this phase, the system is fully deployable, defensible, and ready for evaluation.

1.4 Technical Stack & Architecture

The system is developed using a modern, scalable, and enterprise-ready technology stack designed to support extensibility, security, and high performance:

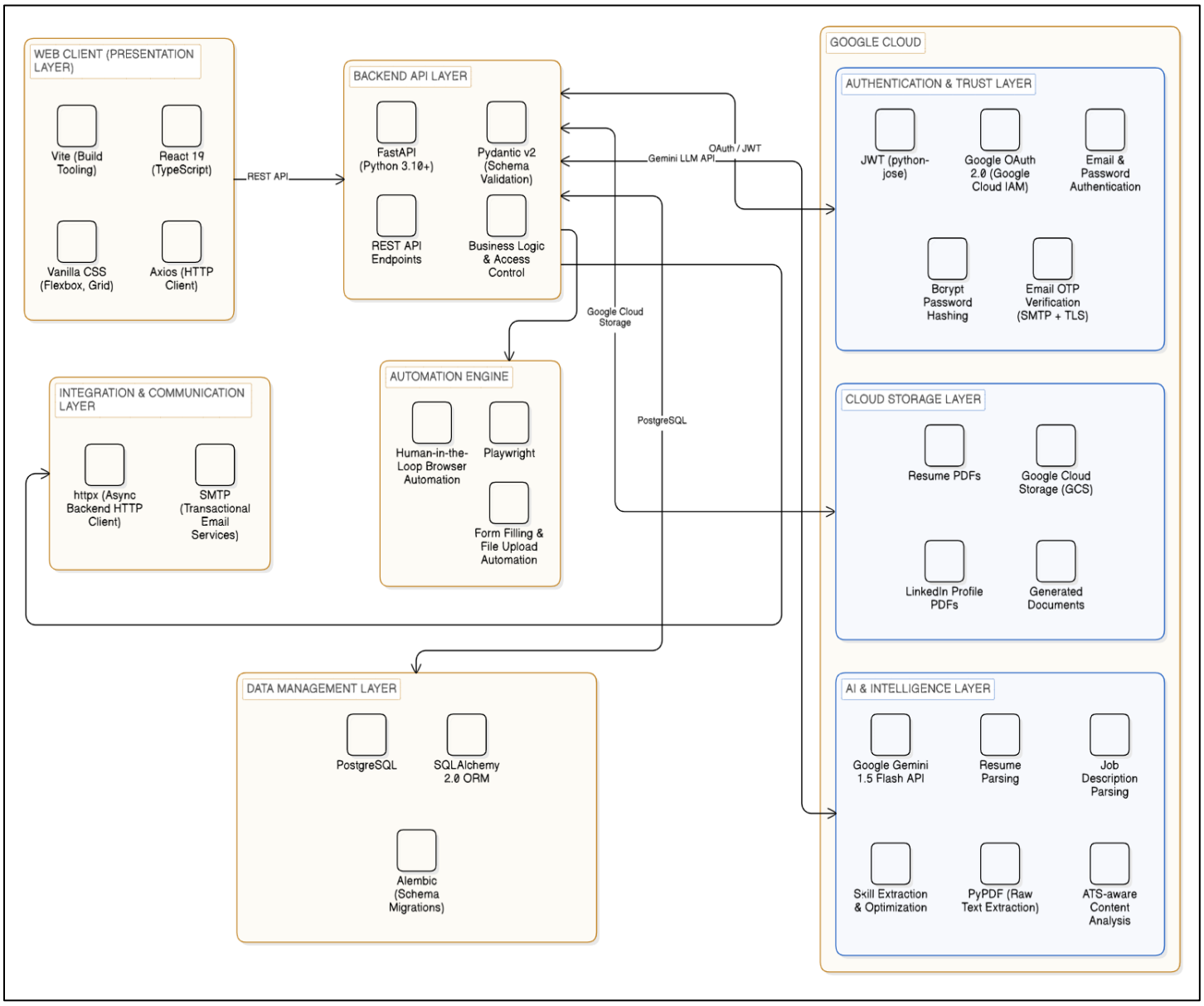

Figure 2: Illustration of Layered Technology Stack

- Backend Framework: Fast API (Python 3.10+) with Pydantic v2 for strict schema validation.
- Frontend Framework: React 19 with TypeScript, using Vite for rapid development and optimized builds.
- Database & ORM: PostgreSQL with SQLAlchemy 2.0 and Alembic for schema migrations.

- AI & LLM Processing: Google Gemini 1.5 Flash for context-aware document parsing and skill mapping.
- Authentication & Security: Google OAuth 2.0, JWT-based sessions (python-jose), and bcrypt-based password hashing.
- Document Engineering: PyPDF for structured extraction of raw text from resumes and LinkedIn PDFs.
- Network & API Communication: Axios (frontend) and httpx (backend) for efficient HTTP interactions.
- Email Services: SMTP with TLS encryption for OTP verification and transactional notifications.
- Styling: Vanilla CSS using modern layout techniques such as Flexbox and Grid.
- Automation Engine: Playwright, designed for controlled, human-in-the-loop browser automation.

1.5 Current Project Progress Status

This section presents the current development progress of the AI-Powered Career ERP Platform, mapped against the planned system roadmap. The objective of this section is to transparently communicate which components have been successfully implemented, which are under active development, and which are planned for subsequent phases. This structured progress tracking demonstrates disciplined execution and alignment with enterprise software engineering practices.

Phase	Phase Title	Objective	Key Activities Covered	Status
Phase 1	Authentication & Trust Layer	Establish secure identity and access control	User schema design, email-password auth, bcrypt hashing, JWT sessions, Google OAuth 2.0, email OTP verification, protected routes, auth UI	Completed
Phase 2	ERP-Style Career Profile Core	Build unified career identity as single source of truth	Versioned profile schema, mandatory vs optional fields, manual entry, resume upload, LinkedIn PDF ingestion, AI parsing, profile review, completion logic	Completed
Phase 3	Dashboard & User Experience Control	Provide visibility and control to users	Active profile retrieval, profile completion computation, last updated timestamp, edit profile flow, logout, profile version updates	Completed
Phase 4-8	Planned			Planned

Table 1: Progress Table till Phase 1 of Evaluation

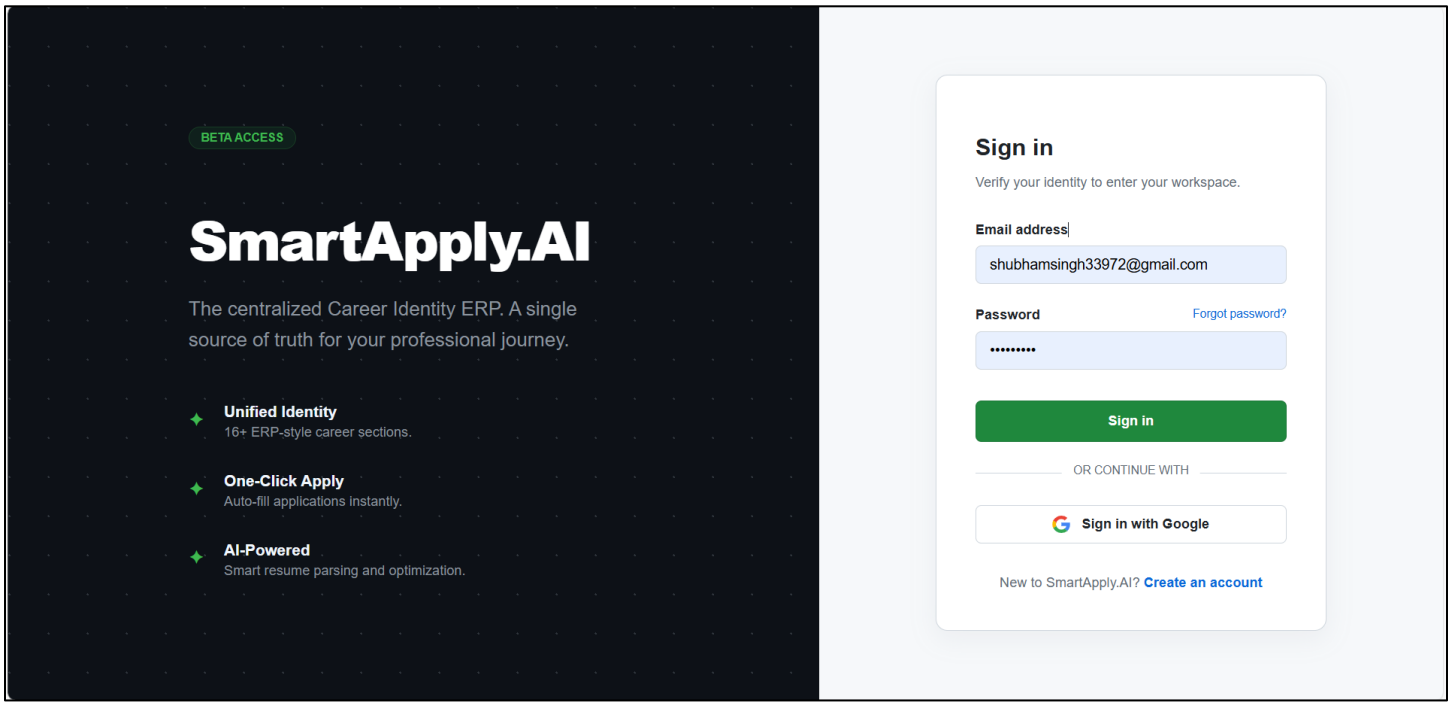

Figure 3: Landing Page/Login Page

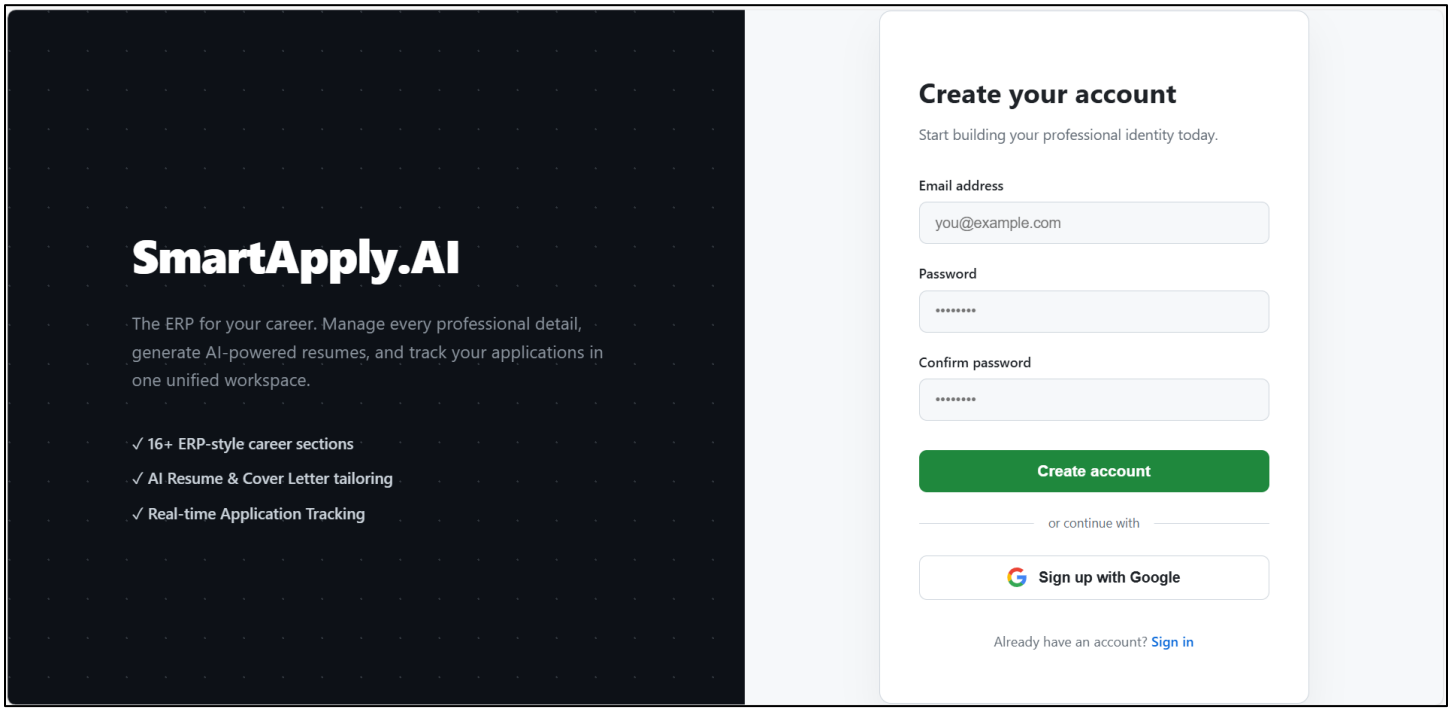

Figure 4: Register Page

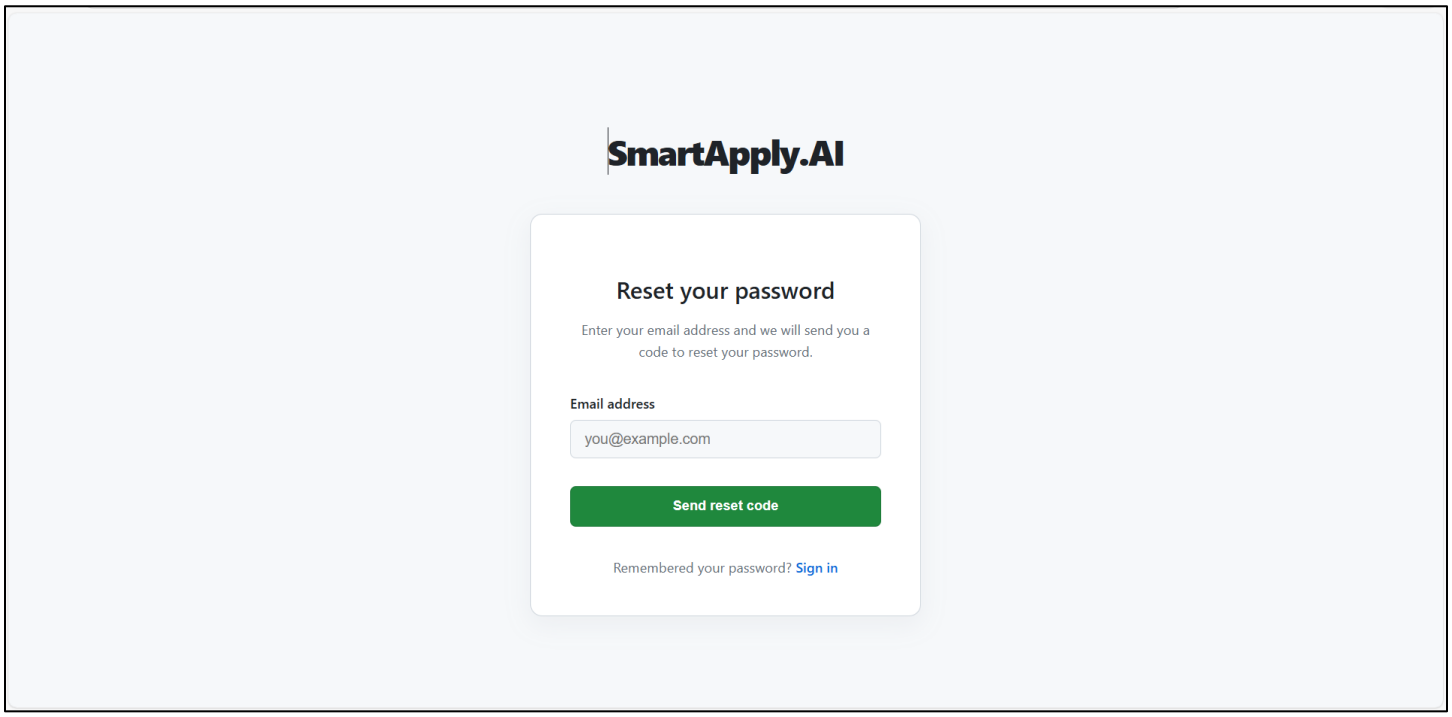

Figure 5: Forgot Password Page

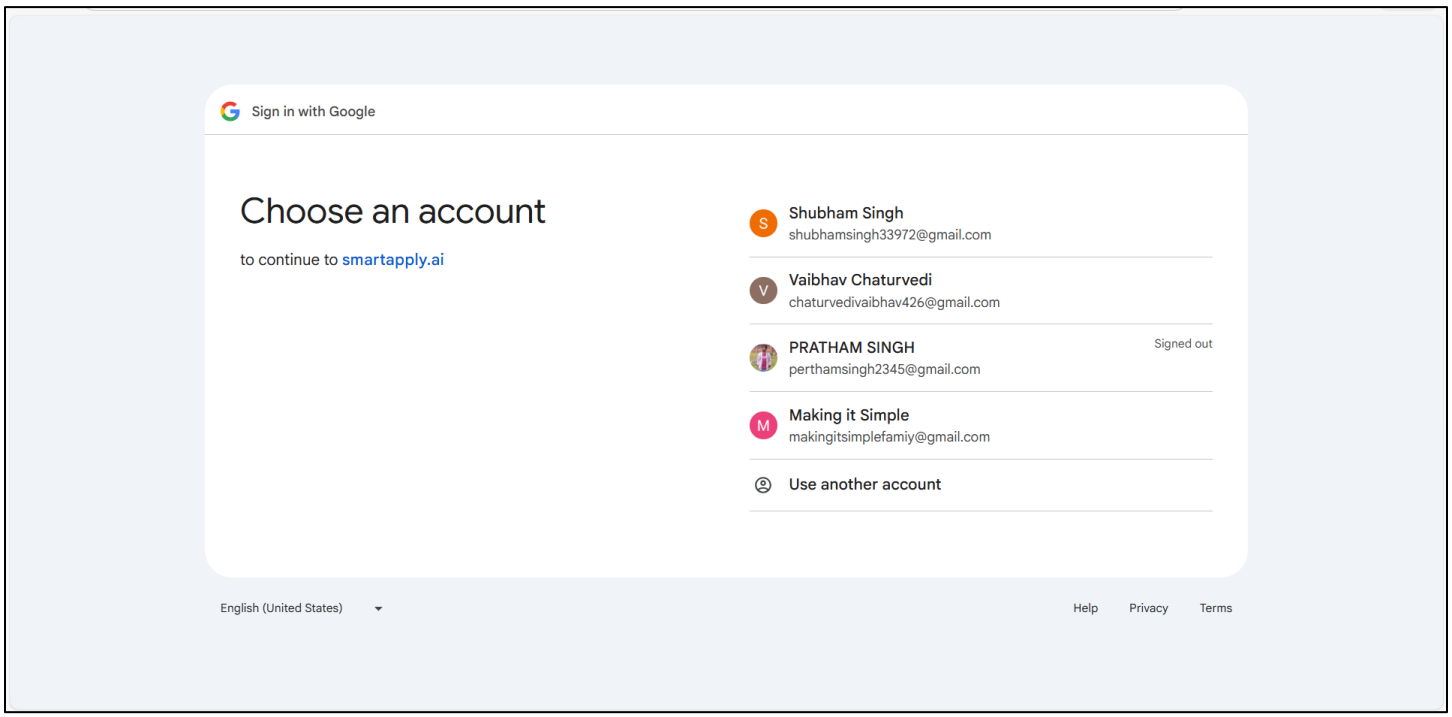

Figure 6: Sigin/Signup using Gmail (OAuth)

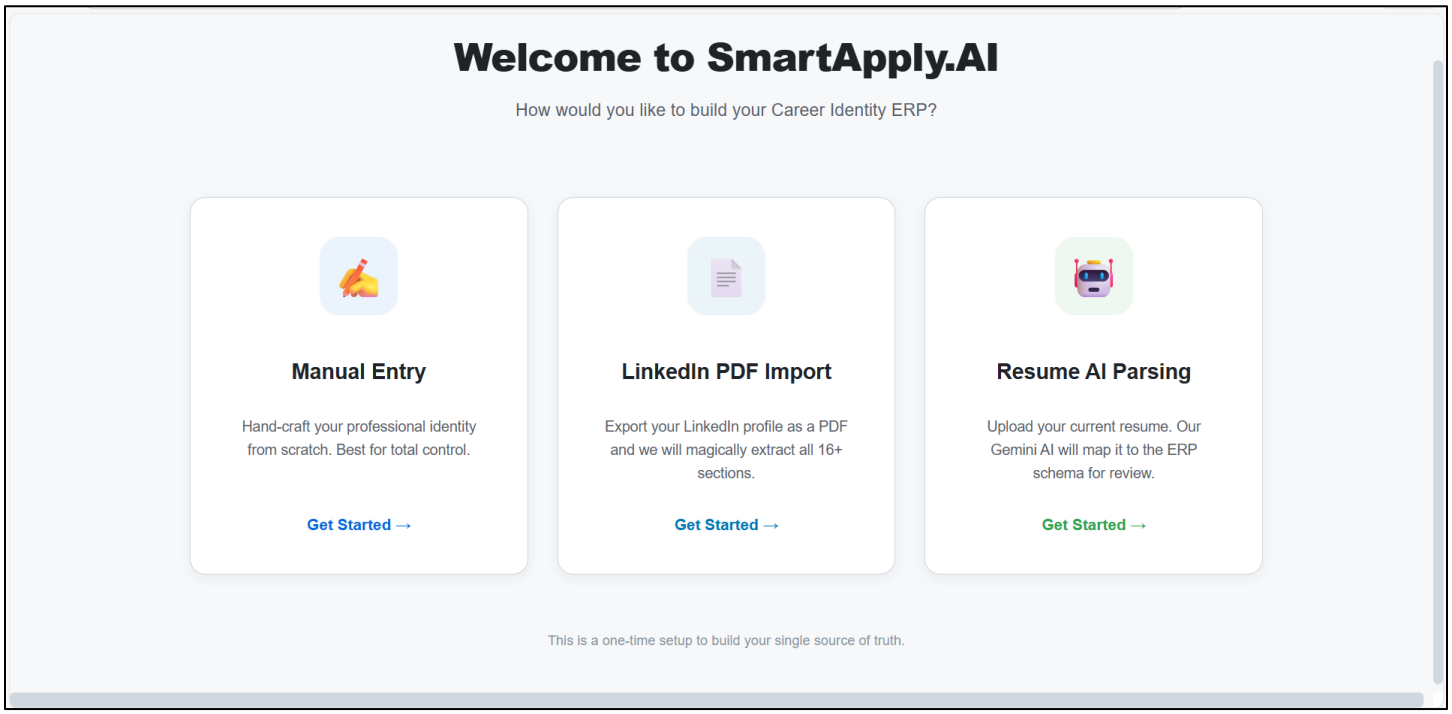

Figure 7: Onboarding Page

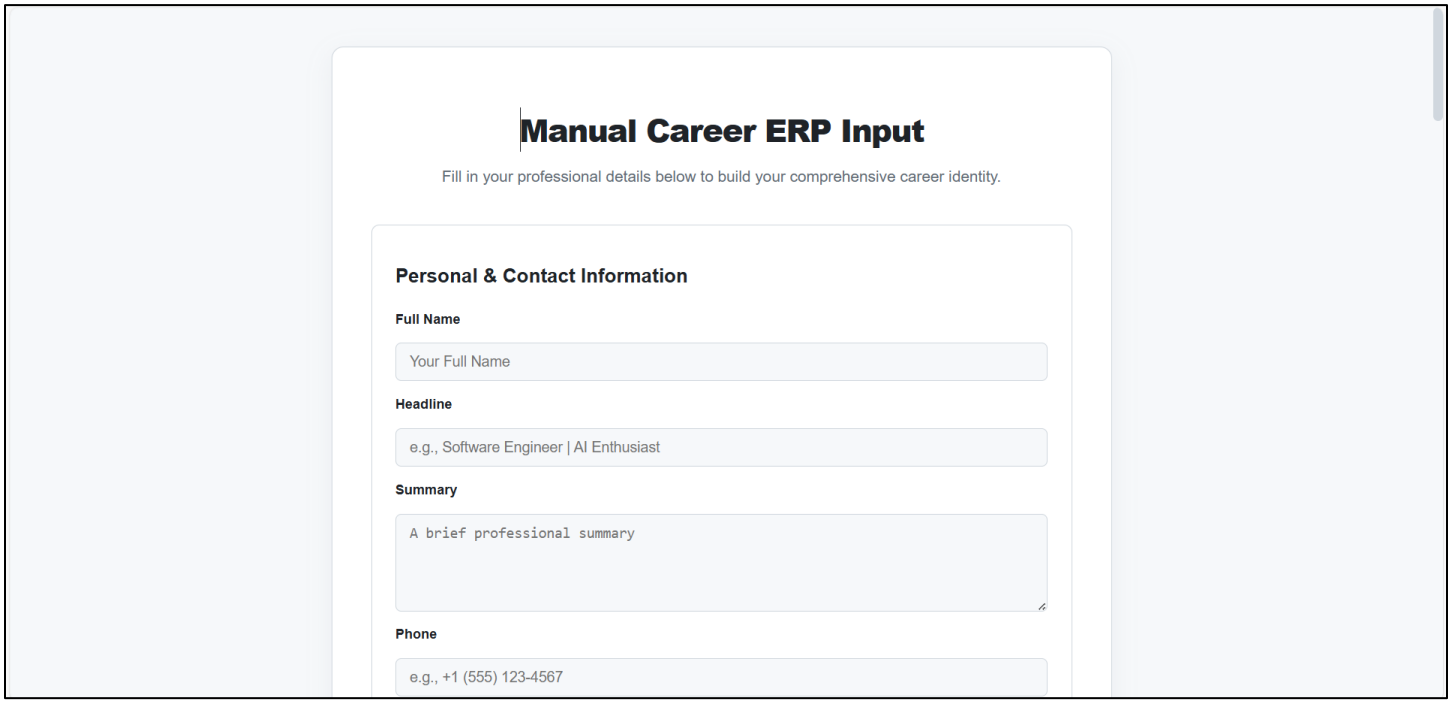

Figure 8: Manual Entry (Onboarding)

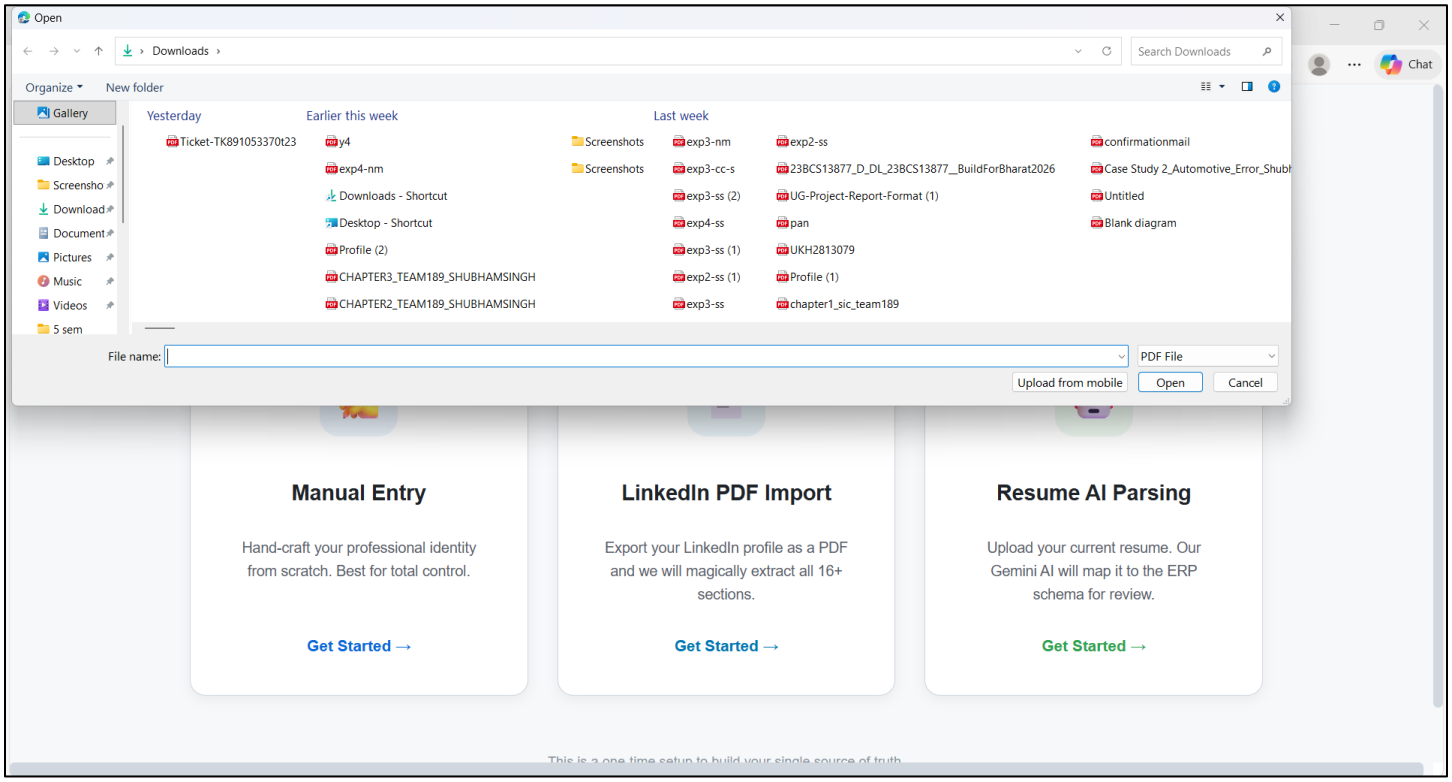

Figure 9: LinkedIn/Resume PDFs upload option for parsing for onboarding

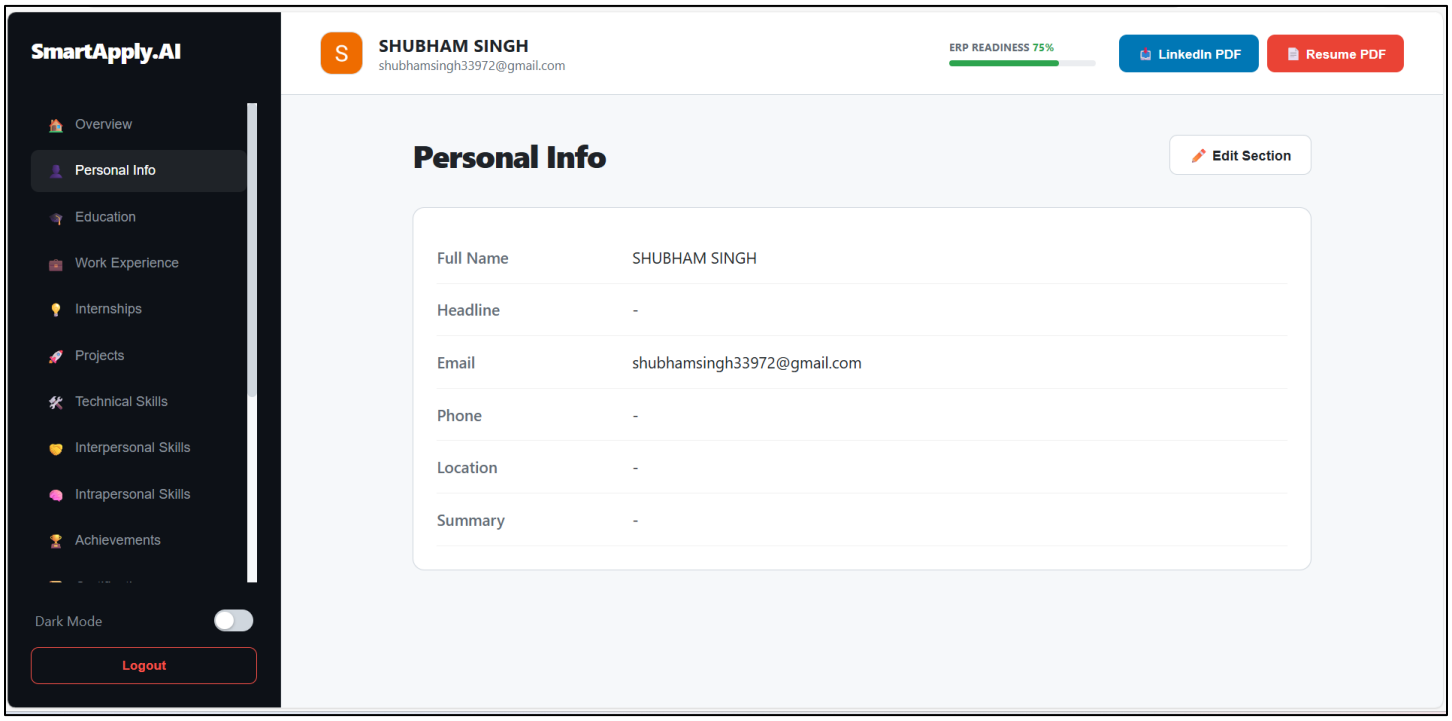

Figure 10: Dashboard Page

1.6 Organization of Report

This report is organized into five chapters to systematically present the problem, methodology, implementation, and outcomes of the proposed system.

Chapter 1: Introduction presents the overall background of the project, identifies the problem and client needs, outlines the objectives, and provides an overview of the proposed solution along with the development plan and current progress.

Chapter 2: Literature Review / Background Study examines existing research, technologies, and systems related to recruitment platforms, Applicant Tracking Systems, AI-based resume analysis, and automation tools. This chapter highlights the limitations of current approaches and identifies the research and technical gaps addressed by the proposed system.

Chapter 3: Design Flow / Process describes the system architecture, workflow design, data flow, and methodological approach adopted in developing the platform. It explains how different components such as authentication, career ERP, AI intelligence, and automation interact within the system.

Chapter 4: Results, Analysis, and Validation presents the outcomes of the implemented system, including performance observations, validation results, and comparative analysis. This chapter evaluates the effectiveness of the proposed solution in addressing the identified problem.

Chapter 5: Conclusion and Future Work summarizes the contributions and key findings of the project and discusses potential enhancements and extensions that can be explored in future iterations of the system.

CHAPTER 2. LITERATURE REVIEW / BACKGROUND STUDY

This chapter presents a focused literature review on digital recruitment systems, AI-based resume analysis, and automated job matching technologies. It examines the evolution of recruitment practices, evaluates existing ATS and AI-driven hiring tools, and identifies their limitations. The review highlights key research gaps, including career data fragmentation, lack of explainable job matching, and absence of unified identity management frameworks. These findings establish the foundation for the proposed AI-powered Career ERP platform and clarify its problem statement and objectives.

2.1 Evolution of Digital Recruitment Systems (Timeline Analysis)

The recruitment ecosystem has undergone profound transformation over the past three decades, evolving from manual paper-based processes to digitally integrated, AI-enabled, and increasingly automated hiring systems. As global job markets expanded and internet penetration increased, organizations adopted technology-driven recruitment mechanisms to handle growing applicant volumes efficiently. Understanding this historical evolution is critical to contextualizing present challenges such as career data fragmentation, algorithmic filtering, lack of explainability, and ethical automation concerns.

2.1.1 1990s: Paper-Based Screening Era

During the 1990s, recruitment was predominantly manual and localized. Candidates submitted printed resumes through postal mail or in person, and hiring managers physically screened applications. Recruitment advertisements were primarily published in newspapers and magazines.
Although this approach ensured direct human evaluation, it suffered from significant limitations including:

- Geographic constraints
- Slow processing time
- Limited applicant reach
- High operational effort

This era represents the pre-digital stage of recruitment, where scalability was inherently restricted.

2.1.2 Early 2000s: Rise of Online Job Portals

With the rapid expansion of internet access in the early 2000s, recruitment entered its first major digital phase. Online job portals such as Monster, CareerBuilder, HotJobs, and Indeed enabled electronic job postings and online resume submissions.
Major financial milestones from this period demonstrate the scale of digital adoption:

- Yahoo acquired HotJobs in 2002 for approximately $436 million.
- CareerBuilder reached a valuation of nearly $2 billion by 2006.
- Monster Worldwide reported peak revenues of approximately $1.3 billion in 2008.

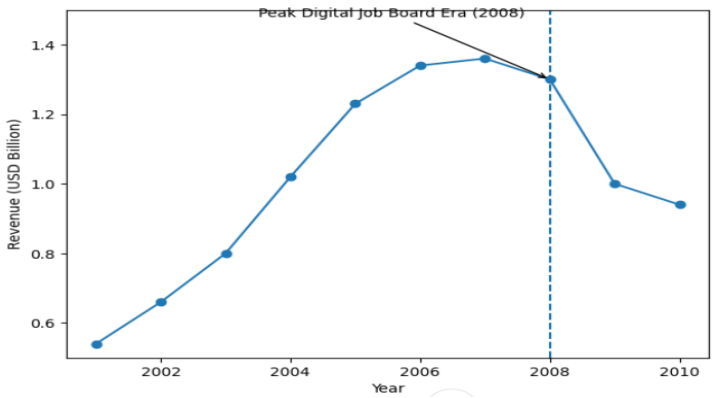

Figure 11: Revenue Growth of Monster Worldwide (2001–2010)

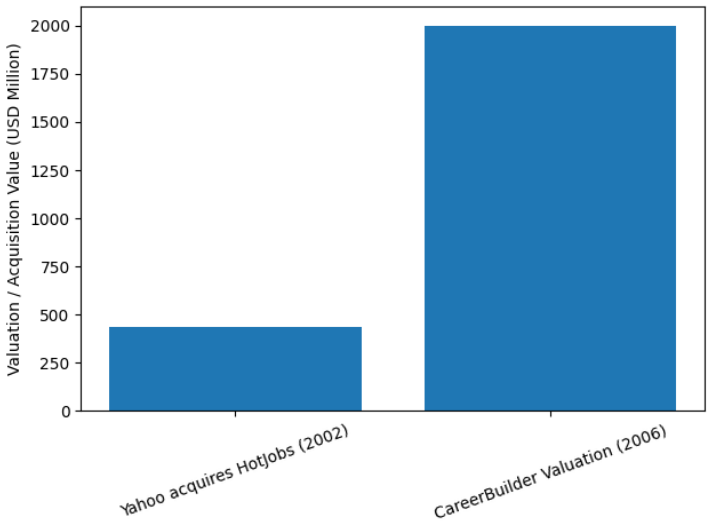

Figure 12: Major Financial Milestones in Early Online Job Platforms (HotJobs Acquisition & CareerBuilder Valuation)

The rapid adoption of digital job portals significantly increased applicant volumes, which eventually made manual screening unsustainable. This set the stage for automated filtering systems.

2.1.3 2010–2015: Emergence and Expansion of ATS Automation

As application volumes surged, organizations required scalable mechanisms to process large candidate pools efficiently. This led to widespread adoption of Applicant Tracking Systems (ATS) during the early 2010s.

ATS platforms automated:

- Resume collection
- Keyword-based filtering
- Structured application parsing
- Candidate ranking

Industry reports indicate that a majority of medium and large enterprises adopted ATS platforms during this period, with particularly high penetration among Fortune 500 companies. Research also suggests that approximately 75% of resumes are filtered before reaching human recruiters.

While ATS systems improved operational efficiency, they introduced new challenges:

- Heavy reliance on keyword matching
- Formatting sensitivity
- Reduced contextual evaluation
- Increased automation bias

This marked the transition from digital recruitment to automated recruitment.

2.1.4 2015–2020: AI-Driven Recruitment and Predictive Hiring

Between 2015 and 2020, recruitment systems evolved beyond rule-based filtering into AI-enabled parsing and predictive analytics. Advances in Natural Language Processing (NLP) allowed platforms to extract structured information from unstructured resumes, including skills, certifications, and work experience.

Machine learning models began assessing candidate-job compatibility using historical hiring data and skill similarity metrics. This shift introduced more data-driven hiring approaches.

However, significant limitations remained:

- Black-box scoring mechanisms
- Limited explainability
- Bias concerns
- Fragmented candidate identity data

Although AI enhanced efficiency, transparency and fairness challenges persisted.

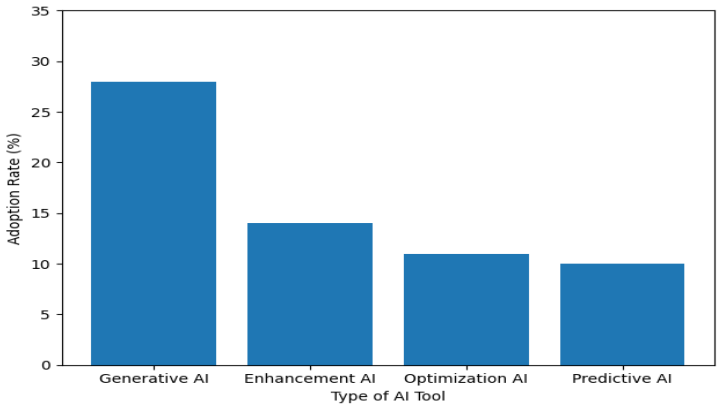

Figure 13: AI tool adoption among HR leaders (SHRM Current Events Pulse Survey, August 2024), showing generative AI as the most widely implemented AI technology.

SHRM's August 2024 survey indicates that generative AI tools represent the most widely adopted form of artificial intelligence in HR functions, with 28% of HR leaders reporting active implementation. In contrast, other AI categories such as enhancement (14%), optimization (11%), and predictive AI (10%) demonstrate significantly lower adoption rates. These findings suggest that while AI integration in HR is progressing, adoption remains in early stages. This reflects the transitional nature of the current recruitment landscape, where LLM-powered tools are emerging but not yet universally institutionalized.

2.1.5 2020–2022: Digital Acceleration and Remote Recruitment Transformation

The global shift toward remote work accelerated digital recruitment adoption. Organizations increasingly relied on:

- Online application platforms
- Virtual interviews
- Automated screening tools
- Remote onboarding systems

Geographical boundaries diminished, expanding talent pools globally. Application volumes further increased, intensifying reliance on automation.

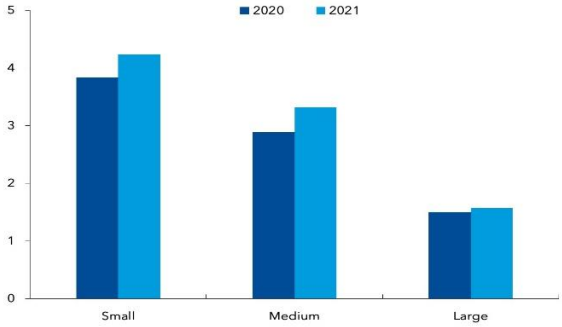

Figure 14: Share of Workers with Internet Access (Source: Eurostat and IMF Staff Calculations)

2.1.6 2022–Present: Large Language Model (LLM) and Generative AI Era

The introduction of Large Language Models in 2022 significantly transformed recruitment technologies. Unlike earlier AI systems that relied primarily on statistical modeling, LLM-based systems offer contextual reasoning and generative capabilities.

Modern recruitment systems now leverage LLMs for:

- Resume rewriting and optimization
- Job description analysis
- Skill gap identification
- Personalized cover letter generation
- Context-aware candidate matching

Recent survey data from SHRM (2024) indicates that generative AI tools are currently the most widely adopted AI technology within HR functions.

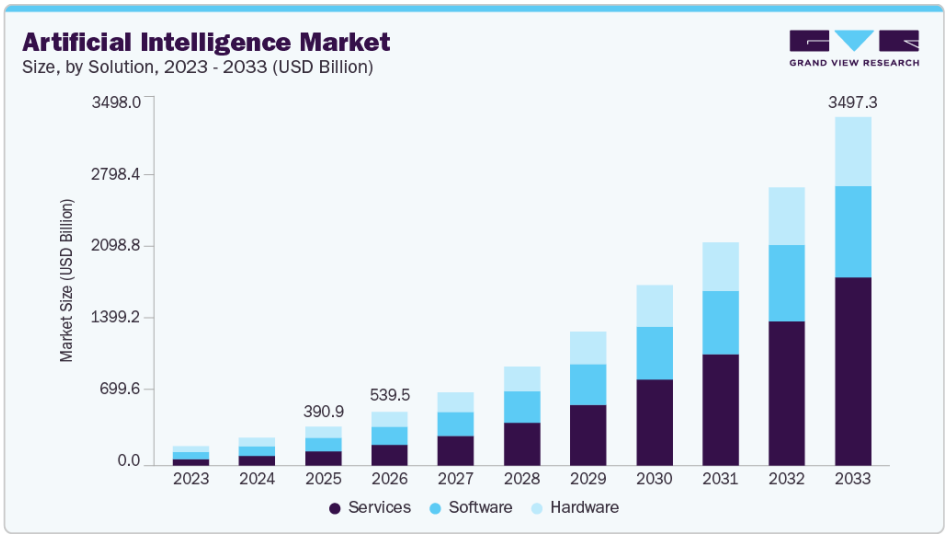

Figure 15: AI Market (2023-2033)

Despite advancements, systemic challenges remain:

- No unified, version-controlled professional identity
- Lack of explainable relevance scoring
- Ethical concerns around blind automation
- Fragmented candidate data ecosystems

Evolution Summary

The evolution of recruitment systems can be summarized as follows:

Manual Screening → Digital Job Portals → ATS Automation → AI Parsing → LLM Semantic Intelligence

Each stage improved scalability and automation. However, none fully addressed the need for a unified, candidate-centric professional identity framework with explainable AI reasoning and ethical automation controls.

This persistent gap provides the foundation for the proposed AI-powered Career ERP platform.

2.2 Bibliometric Analysis of Related Research

To understand the academic progression of AI in recruitment, a bibliometric analysis of research trends was conducted focusing on resume parsing, AI-based hiring systems, candidate-job matching algorithms, and explainable AI in HR.

Research publications in AI-driven recruitment increased significantly after 2016, corresponding with advances in NLP and machine learning. A marked spike in research activity occurred post-2022 following the emergence of generative AI and LLM technologies.

Common research keywords include:

- Resume Parsing using NLP
- Skill Extraction
- Candidate–Job Matching
- Predictive Hiring
- Explainable AI in HR
- Automated Recruitment Systems

While research heavily emphasizes automation and AI accuracy, limited work addresses unified professional identity management and integrated ERP-style frameworks.

2.3 Review of Existing Systems and Platforms

Existing recruitment systems address specific components of hiring but lack holistic integration.

Platform / Category	Strength	Limitation
LinkedIn	Professional networking & visibility	Limited role-specific optimization
Resume Builders	Template formatting & structure	No intelligent job matching
ATS Platforms	Employer-side filtering & automation	Not candidate-centric
AI Resume Tools	Keyword optimization	Limited explainability

Although these systems improve certain aspects of recruitment, they do not provide a unified, version-controlled, candidate-owned identity framework.

2.4 Summary of Research Gaps

Research Area	Existing Approach	Identified Gap
Resume Parsing	NLP extraction	Limited explainability
Job Matching	Keyword overlap	Weak contextual reasoning
ATS Systems	Employer-centric filtering	No candidate-owned identity
Automation	Script-based automation	Lack of human-in-the-loop control
Resume Platforms	Static document generation	No unified ERP backbone

There is no unified system that integrates identity management, AI intelligence, explainable scoring, and ethical automation within a single candidate-centric architecture.

2.5 Linking Literature Review to the Proposed System

The proposed AI-powered Career ERP platform addresses the identified gaps by integrating:

- Structured and version-controlled professional identity modeling
- AI-driven resume and job description parsing
- Explainable relevance scoring mechanisms
- Human-in-the-loop browser automation
- Ethical transparency and audit logging

Unlike traditional resume tools or ATS platforms, this system treats professional identity as a reusable and evolving digital asset rather than a static document.

2.6 Refined Problem Definition

The absence of an interoperable, intelligent, and candidate-centric professional identity management system leads to inefficient job applications, poor skill-role alignment, automation bias, and reduced trust in AI-driven recruitment processes.

2.7 Goals and Objectives

General Goal

To design and develop an AI-powered, ERP-style career platform that intelligently manages professional identity and enhances job application relevance.

Specific Objectives

- Develop secure authentication and identity management
- Build unified, version-controlled career profile architecture
- Implement AI-based resume and job description parsing
- Design explainable job relevance scoring
- Enable controlled, human-in-the-loop automation
- Ensure ethical transparency and deployment readiness